characterize a particular abnormality is called a syndrome. In humans, Down's syndrome, Turner's syndrome, Klinefelter's syndrome, Patau's syndrome are some of the examples of chromosomal disorders.

a. Autosomal aneuploidy in human beings

Several autosomal aneuploidies have been reported in human beings. eg. Down's syndrome (21-Trisomy), Patau's syndrome (13-Trisomy).

1. Down's Syndrome/Trisomy – 21

Trisomic condition of chromosome - 21 results in Down's syndrome. It is characterized by severe mental retardation, defective development of the central nervous system, increased separation between the eyes, flattened nose, ears are malformed, mouth is constantly open and the tongue protrudes.

2. Patau's Syndrome/Trisomy-13

Trisomic condition of chromosome 13 results in Patau's syndrome. Meiotic non disjunction is thought to be the cause for this chromosomal abnormality. It is characterized by multiple and severe body malformations as well as profound mental deficiency. Small head with small eyes, cleft palate, malformation of the brain and internal organs are some of the symptoms of this syndrome.

b. Allosomal abnormalities in human beings

Mitotic or meiotic non-disjunction of sex chromosomes causes allosomal abnormalities. Several sex chromosomal abnormalities have been detected. Eg. Klinefelter's syndrome and Turner's syndrome.

1. Klinefelter's Syndrome (XXY Males)

This genetic disorder is due to the presence of an additional copy of the X chromosome resulting in a karyotype of 47,XXY. Persons with this syndrome have 47 chromosomes (44AA+XXY). They are usually sterile males, tall, obese, with long limbs, high pitched voice, under developed genitalia and have feeble breast (gynaecomastia) development.

2. Turner's Syndrome (XO Females)

This genetic disorder is due to the loss of a X chromosome resulting in a karyotype of 45,X. Persons with this syndrome have 45 chromosomes (44 autosomes and one X chromosome) (44AA+XO) and are sterile females. Low stature, webbed neck, under developed breast, rudimentary gonads lack of menstrual cycle during puberty, are the main symptoms of this syndrome.

4.10 Extra chromosomal / Cytoplasmic inheritance



Certain characters are controlled by non- nuclear genomes found in chloroplast, mitochondria, infective agents and plasmids. These characters do not reveal Mendelian pattern of inheritance. The inheritance of the extra chromosomal genes are found to exhibit maternal influence. Maternal effect is due to the asymmetric contribution of the female parent to the development of zygote. Although both male and female parents contribute equally to the zygote in terms of chromosomal genes, the female parent usually contributes the zygote's initial cytoplasm and organelles, since the sperms contain very little cytoplasm. If there are hereditary units in the cytoplasm, these will be transmitted to the offsprings through the egg, so the offsprings exhibit maternal effect.

The cytoplasmic extranuclear genes have a characteristic pattern of inheritance which do not resemble the genes of nuclear chromosomes and is known as extra chromosomal or extra nuclear or cytoplasmic inheritance and exhibit maternal influence. In extra nuclear inheritance, male and female parents contribute equally their nuclear genes to the progeny but do not make equal contribution of extra chromosomal genes hence, the crosses can yield different (or) non Mendelian results. Cytoplasmic inheritance

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